Kushal Dudipala

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Education

2022-2026 **Georgia Institute of Technology**
B.S. in Computer Science, Concentration in Intelligence & Theory (Highest Honors) GPA: 3.7/4.0
B.S. in Biology (Highest Honors) GPA: 3.6/4.0

Research Experience

Lab Manager at Murty Lab, *Georgia Tech* Jun 2026 - present
Current lab manager at Murty lab, working on neural-based models and studying atypical neural behavior.
Outcomes: 1 first-author workshop paper (1 related in-prep journal paper), 1 first-author in-prep journal paper

Undergraduate Research Assistant at Murty Lab, *Georgia Tech* Aug 2024 - Jun 2026
Worked on writing libraries, topographic modeling, and high-throughput encoding models
Outcomes: 1 submitted co-author journal paper, multiple conference presentations

Undergraduate Researcher at Najafi Lab, *Georgia Tech* Aug 2023 - May 2024
Trained + built a high-throughput pipeline for deep neural networks on video for pose estimation.
Outcomes: 1 co-author on conference presentations

Featured Works

Standardized human tests show that most vision models are face-blind Dec 2026
(*NeurIPS 2026 Workshop Interp4Discovery*)
Kushal Dudipala, Florencia Martinez-Addiego, Dobromir Rahnev, Bradley Duchaine, N. Apurva Ratan Murty
Administering standardized human face-recognition tests to a diverse axis of ANNs, we found most models are 'face-blind' and even the best fall short of human super-recognizers.

Zero-shot evaluations expose structured generalization limits in predictive models of the brain May 2027
(*bioRxiv, Under review at Nature Neuroscience*)
Ruolin Wang, Mayukh Deb, Alex Abate, Alish Dipani, **Kushal Dudipala**, Sanjana Chillarege, Kruthik Ravikanti, Yuxuan Li, Haider Al-Tahan, Ranjani Koushik, Elizabeth Mieczkowski, Ho Yin Fung, Nancy Kanwisher, N. Apurva Ratan Murty
Built a zero-shot framework that tests frozen ANN models of human visual cortex on new fMRI datasets and classic cognitive experiments, revealing generalization failures that standard cross-validation hides.

Conferences

† *indicating mentee*

Kushal Dudipala, Mayukh Deb, Ben Lahner, and Ratan Murty. "Linking neural representations to behavior with a unified model of human vision". In: *Vision Sciences Society 2026 Annual Meeting*. Poster presentation 23.349. Banyan Breezeway, May 2026. URL: `https://www.visionsciences.org/presentation/?id=4401`.

Navneet Prakash†, Mayukh Deb, **Kushal Dudipala**, and Apurva Ratan Murty. "Topographic constraints drive the emergence of grid-cell center-surround connectivity in path-integrating networks". In: *9th Annual Conference on Cognitive Computational Neuroscience*. 2026.

Ruolin Wang, Yuxuan Li, Mayukh Deb, **Kushal Dudipala**, Kruthik Ravikanti, Sanjana Chillarege, Arya Bhanushali, Ranjani Koushik, Aashraya Katiyar, and N Apurva Ratan Murty. "Cortex-Canvas: An Interactive Web Interface for Executing and Evaluating Models of Category-Selective Regions in Human Visual Cortex". In: *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*. 2026, pp. 1–7.

Kushal Dudipala, Mayukh Deb, Benjamin Lahner, and N. Apurva Ratan Murty. "Neuromap: Bridging Neural Activity and Perceptual Behavior Through Joint Optimization". In: *Biology Capstone Symposium*. Poster presentation. Georgia Institute of Technology, 2024.

Kushal Dudipala, Mayukh Deb, Benjamin Lahner, and N. Apurva Ratan Murty. "Neuromap: Bridging Neural Activity and Perceptual Behavior Through Joint Optimization". In: *Neuroscience Undergraduate Symposium*. Poster presentation. Georgia Institute of Technology, 2024.

Tim Stamm, **Kushal Dudipala**, Farzaneh Najafi, et al. "Investigating Predictive Processing in Cortico-Cerebellar Circuits". In: *Neuronext Launch Symposium*. Poster presentation. Georgia Institute of Technology, 2024.

Talks

May 2026 **Vision Sciences Society, 2026 satellite event (abstract)**
Roulin Wang, Mayukh Deb, ... , **Kushal Dudipala**, ..., Nancy Kanwisher, N. Apurva Ratan Murty

Science Communications

Kushal Dudipala. "DeepSeek Rattles U.S. Markets". In: *The Technique (Georgia Institute of Technology student newspaper)* (Apr. 2025). News. URL: `https://nique.net/news/2025/04/08/deepseek-rattles-u-s-markets/`.

Kushal Dudipala and Samay Rapelly. "Okefenokee Expansion". In: *The Technique (Georgia Institute of Technology student newspaper)* (2025). News. URL: `https://nique.net/news/okefenokee-expansion`.

Additional experience

AI Safety Fellow @ *GT AI Safety Club* 2025 - 2026

Selected for a cohort-based fellowship covering core topics in AI alignment and safety (interpretability, RLHF, evaluation), with readings, discussions, and a capstone project.

Software Engineering Intern II @ *Elevance Health* May 2024 - August 2024

Independently scoped and built ReconcileDB, a full-stack tool (Python, MongoDB, ReactJS) for detecting inconsistencies across large HIPAA-protected databases, saving 500 work-hours in the migration pipeline.

Awards

NeurIPS 2026 travel award	2026
Georgia Tech Fast-Track to Research Scholarship	2024
Georgia Tech Dean & Faculty Honors List	2022 - 2026
Zell Miller Scholarship	2022 - 2026